Day 20: Inference for logistic regression

Last time we introduced logistic regression and learned to interpret the slope. Today:

1. Logistic regression as a **generalized linear model**
2. **Inference** for a single coefficient (Wald tests and confidence intervals)
3. **Checking conditions** with empirical logit plots and Pearson residuals

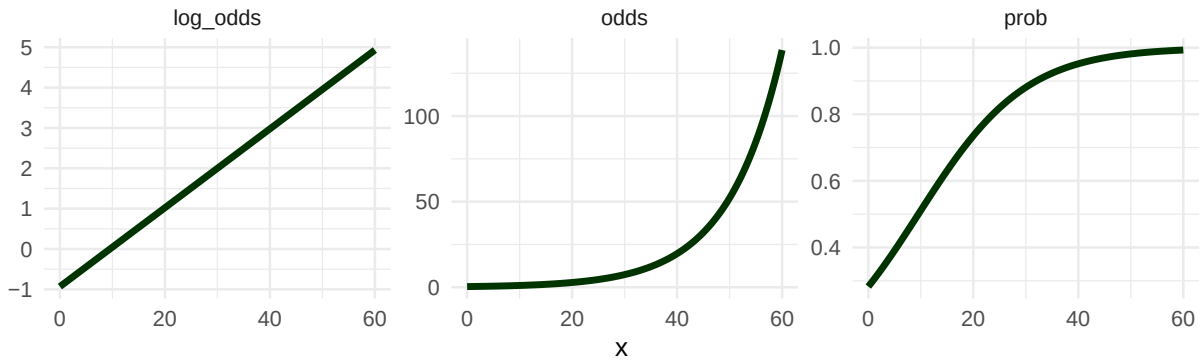
Running example: Fake News. Our goal is to explain whether an article is fake (`type_num` = 1) using

- `title_words`: number of words in the title
- `title_has_excl`: whether the title has an exclamation point
- `title_caps_percent`: percent of the title in capital letters

term	estimate	std. error	z	p-value
Intercept	-2.232	0.672	-3.32	<0.001
title_words	0.120	0.058	2.07	0.038
title_has_exclTRUE	1.998	0.849	2.35	0.019
title_caps_percent	0.050	0.029	1.69	0.091

Warm up: Write the fitted model on the log-odds scale.

Fitted regression model: 3 scales



Generalized linear models

Both linear regression and logistic regression have three parts:

- **Linear predictor:** $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$
- **Link function:** connects η to the mean of Y, $\mu = g(\eta)$
- **Response distribution:** the distribution of Y with mean μ

How do we estimate the coefficients?

Maximum likelihood (ML) estimation

In linear regression we found the $\hat{\beta}$'s that minimized the sum of squared errors. The coefficients in logistic regression are estimated by finding the $\hat{\beta}_0, \ldots, \hat{\beta}_p$ that maximize the probability of the observed outcomes (called the *Likelihood*)

$$L(\beta) = P(Y_1 = y_1, \ldots Y_n = y_n | \beta, X) = \prod_{i=1}^{n} \pi(X)^{y_i} \Big[1 - \pi(X)\Big]^{1-y_i}$$

where

$$\pi(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

If the sample is large and the model is correct, the ML estimators are

1. essentially unbiased
2. as precise as any other unbiased estimator
3. approximately **normal**

That last fact is what lets us do inference.

Wald test for a coefficient

Hypotheses: H_0: _____ vs. H_a: _____
Test statistic: $z =$ _____
Reference distribution: _____

Wald-based Confidence Interval

A normal-based confidence interval for _____ is:

$$\hat{\beta}_i \pm z^*_{1-\alpha/2} SE(\hat{\beta}_i)$$

CI for the multiplicative effect on _____ of success for a 1- unit change in x (odds ratio of Success for $x + 1$ vs. x):

$$e^{\widehat{\beta}_i - z^*_{1-\alpha/2} SE(\widehat{\beta}_i)} \quad \text{to} \quad e^{\widehat{\beta}_i + z^*_{1-\alpha/2} SE(\widehat{\beta}_i)}$$

for a C-unit change in x:

$$e^{C \cdot \widehat{\beta}_i - z^*_{1-\alpha/2} C \cdot SE(\widehat{\beta}_i)} \quad \text{to} \quad e^{C \cdot \widehat{\beta}_i + z^*_{1-\alpha/2} C \cdot SE(\widehat{\beta}_i)}$$

Example: Is the number of words in the title associated with the odds of being fake news, after accounting for the other variables?

term	estimate	std.error	statistic	p.value
title_words	0.12	0.058	2.072	0.038

Conclusion in context:

Confidence intervals for a coefficient

A normal-based CI for β_i:

$$\hat{\beta}_i \pm z^*_{1-\alpha/2} \cdot SE(\hat{\beta}_i)$$

To get a CI for the **odds ratio**, exponentiate the endpoints:

$$e^{\widehat{\beta}_i - z^* SE(\widehat{\beta}_i)} \quad \text{to} \quad e^{\widehat{\beta}_i + z^* SE(\widehat{\beta}_i)}$$

For a C-unit change in x, multiply the endpoints of the CI for β_i by _____ *before* exponentiating.

Example: 95% CI for the effect of one more word in the title ($z^* = 1.96$).

Interpretation: We are 95% confident that …

Example: What about a 10-word increase in the title?

term	estimate	conf.low	conf.high	or_low	or_high	or_low_10	or_high_10
title_words	0.12	0.008	0.237	1.008	1.268	1.084	10.737

(Note: `tidy(..., conf.int = TRUE)` uses a profile-likelihood interval, so its endpoints differ slightly from the Wald interval above.)

Conditions for logistic regression

For inference to be valid we need

1. **L**inearity: the _____ is a linear function of the predictors
2. **I**ndependence: no pairs, clusters, ordering in time or space
3. **R**andomness: a random sample (or a randomized experiment)

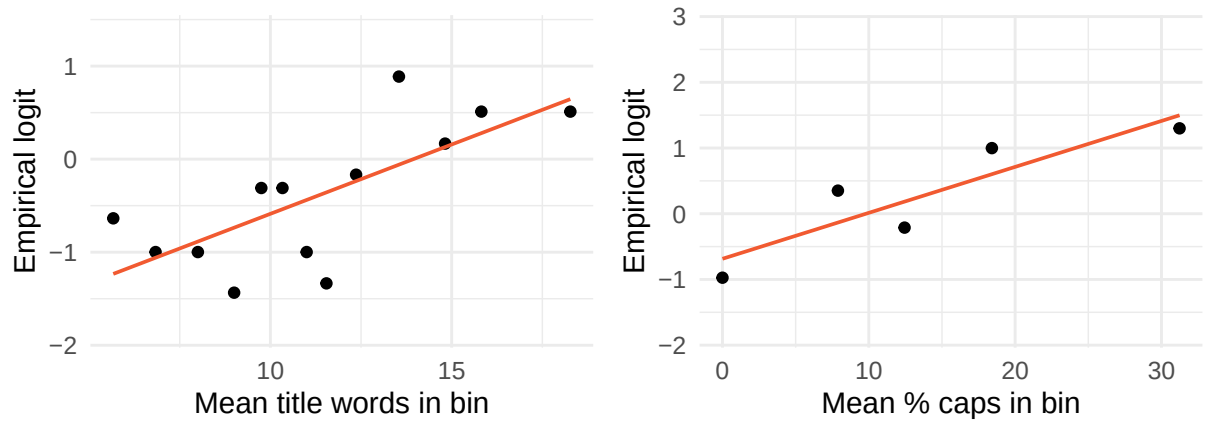
If linearity fails, the model doesn't describe the data. If independence or randomness fail, we can still interpret the coefficients, but our CIs will be too narrow and our p-values too small.

Checking linearity: empirical logits

We can't just plot a 0/1 response against x and look for a line, and ordinary residual plots for binary data are hard to read. Instead we plot the **observed log odds** for groups of similar x values.

1. Divide the observations into g bins of similar x values
2. In each bin, count the successes and failures
3. **Empirical logit** $= \log\left(\dfrac{\#\text{successes} + 0.5}{\#\text{failures} + 0.5}\right)$
4. Plot the empirical logit against the mean of x in each bin and look for a straight line

Why do we add 0.5? _____



Q3. The left plot uses 13 bins of about 11 articles each. Is there evidence that the linearity condition is violated for `title_words`?

Q4. 92 of the 150 titles have 0% capital letters, so we couldn't use equal-size bins for `title_caps_percent` (the right plot uses 5 bins chosen by hand). What do you conclude about linearity?

What to look for:

- A roughly straight trend means linearity is _____.
- A curved trend means we should try _____ (for example, _____).
- Each point is based on few observations, so expect scatter. Look at the overall trend, not at individual points.
- These plots look at one predictor at a time. They don't adjust for the other predictors, which is why we also use Pearson residuals.

Categorical predictors. For `title_has_excl` we calculate the empirical logit in each group:

title_has_excl	n	fake	real	empirical logit
FALSE	132	44	88	-0.69
TRUE	18	16	2	1.89

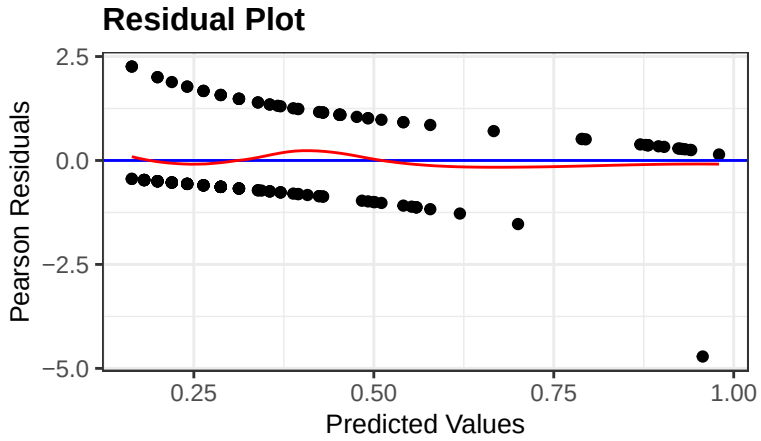
With only two groups, there is no linearity condition to check: _____.

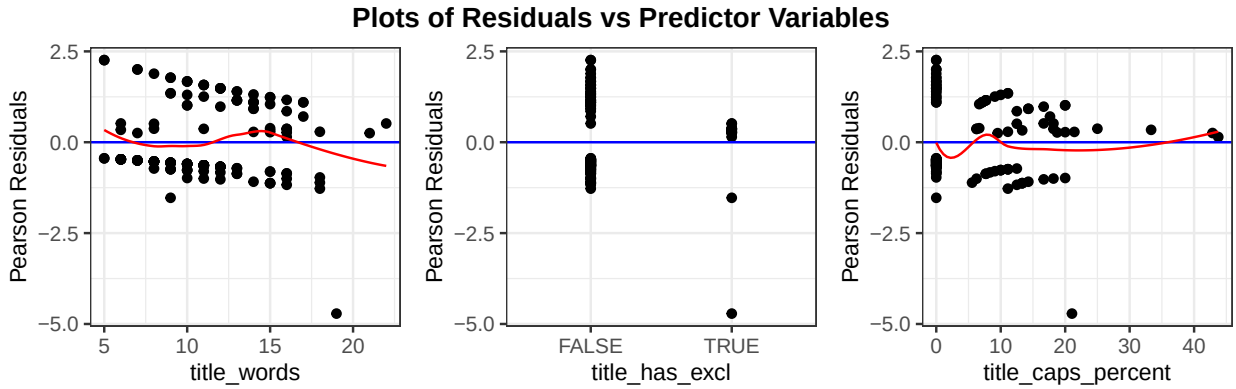
Checking linearity after adjusting: Pearson residuals

The raw difference between the observed outcome and the fitted probability is $y_i - \hat{\pi}_i$. The **Pearson residual** standardizes it:

$$r_{p_i} = \frac{y_i - \hat{\pi}_i}{\sqrt{\hat{\pi}_i(1 - \hat{\pi}_i)}}$$

For binary data the Pearson residuals fall in two curved bands (one for $y = 0$, one for $y = 1$), so the individual points aren't informative. We add a **smoother**:



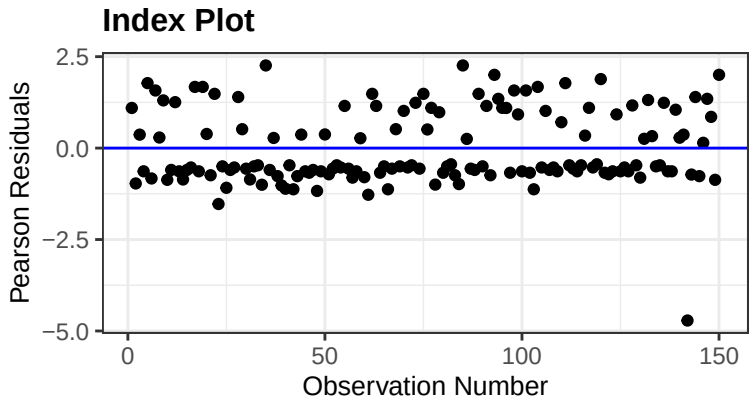


- If the model is adequate, the smoother should be roughly a _____ line at _____.
- There will always be some variation from the horizontal. Smoothers help us notice **major** flaws.
- Pearson residuals for binary data are *not* approximately normal, so we don't use a QQ plot.

Q5. Do the Pearson residual plots agree with the empirical logit plots? Is there evidence of a violation of linearity?

Independence and randomness

- Independence is best assessed from the *description of the data*: Are the observations grouped (e.g., several articles from the same site)? Measured over time? Spatially related?
- An **index plot** of the residuals shows whether the data are ordered in some way (a trend or pattern across the index is a warning sign).
- Randomness: how were the articles chosen? The fake news articles were posted to Facebook in September 2016, in the run-up to the election, so we need to be careful about what population our conclusions apply to.



Q6. What does the index plot suggest about independence? What population do you think we can generalize to?

Summary

Condition	How we check it
Linearity	
Independence	
Randomness	

- Inference for each coefficient uses a _____ test and a _____-based CI.
- Exponentiate the CI endpoints to talk about the **odds ratio**; multiply by C first for a C-unit change.

R function reference

Function	Purpose
`glm(y ~ x, data, family = "binomial")`	Fit a binary logistic regression
`tidy(model, conf.int = TRUE)`	Coefficient table with confidence intervals
`mutate(bin = ntile(x, g))`	Create g equal-size bins of x
`cut(x, breaks)`	Create bins from cut points you choose
`log((successes + 0.5) / (failures + 0.5))`	Empirical logit
`resid_panel(model, plot = "resid", type = "pearson", smoother = TRUE)`	Pearson residual plot with a smoother
`resid_xpanel(model, type = "pearson", smoother = TRUE)`	Pearson residuals against each predictor